iRing: Rich Interaction on Smart Ring with Multi-point Capacitive Touch Screen

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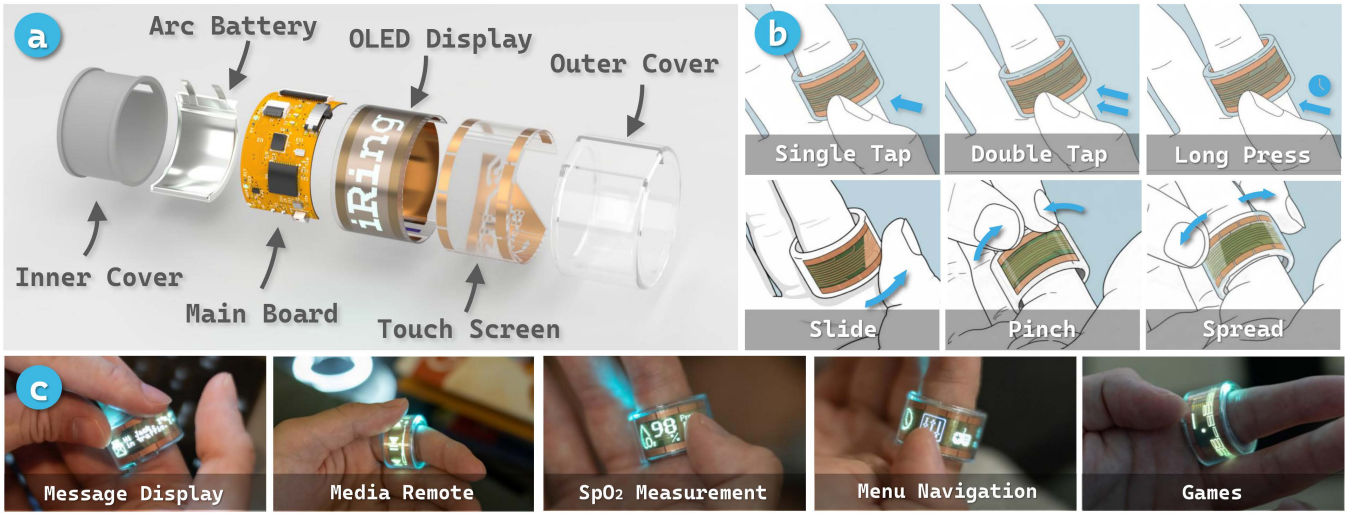


Figure 1: Design and Interaction of iRing. (a) An exploded view of the iRing, including an arc battery, OLED dsplay, outer and inner covers, main board, and a flexible touch screen. (b) Illustration of the six distinct touch primitives supported for direct interaction on the ring's surface. (c) A suite of applications to demonstrate interactive potential: spread on the display to zoom in text, control sound system wirelessly, SpO2 measurement, sliding navigation menus, and interactive games "Breakout".

Abstract

Rising interest in smart rings reflects a growing demand for immediate, lightweight interaction. However, prior designs have largely treated rings as input-only interfaces, or devices with limited on-device visualization, fall short in supporting for tightly coupled, direct, and expressive co-located interaction. To bridge this gap, we present iRing, a lightweight (6.4 g), low-cost ($46.43), open-source, and reproducible smart ring platform centered on a capacitive multi-touch display. We design a compact, multi-touch interface using accessible components that fit the ring's tight form factor. It further integrates a flexible OLED display, BLE micro-controller, IMU, and PPG sensor, supporting both local and remote interaction. To bridge hardware touch sensing and interaction events, we introduce nano interactions—a compact vocabulary that adapts familiar touch primitives for direct, co-located input and visual output on ultra-small, curved surfaces. Nano interactions allow interaction within a single glance–touch cycle, supporting one-handed, low-attention use under motion. We implement a set of applications to explore the design space enabled by multi-touch on a ring. A user study with 12 participants demonstrates reliable touch interaction and provides positive short-term assessments of usability, comfort, and social acceptability, highlighting opportunities for compact interaction surfaces supporting glanceable and on-the-go interaction.

Keywords

Smartring, Wearable, Sensor, Sensing Application, Interaction, Glanceable Display, On-the-go Interaction

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1 Introduction

Smart rings are increasingly attracting attention in the Human Computer Interaction (HCI) domain as compact and lightweight wearable devices that fit naturally into daily life. However, smart ring interactive systems still remain underexplored, offering only partial solutions that fall short of supporting fine-grained, tightly coupled, co-located interaction, constrains the exploration of advanced interaction paradigms. Existing smart rings largely depend on external displays to provide visual feedback, limiting the immediacy and completeness of interaction. As emphasized in foundational HCI principles, such as Nielsen's heuristic of system status visibility and Shneiderman's principle of direct manipulation [37, 48], immediate feedback is essential for effective and satisfying user experiences. Prior works leveraging inertial measurement units (IMUs) for gesture interaction relies on indirect, non-isomorphic mappings between gesture space and interface parameters, increasing cognitive translation cost, meanwhile constrained to unidirectional interaction [38]. Similarly, capacitive proximity sensing approaches are highly susceptible to individual body characteristics and likewise limited to unidirectional interaction [59]. Conversely, interactive ring systems that feature co-located interaction are often constrained by limited design choices. Many rely on simple digital displays paired with single input elements in fixed positions, restricting both the interaction diversity and content richness [27]. Others explore pixel-based displays coupled with motion-based inputs [15, 33, 52], yet these configurations tend to impose steep learning curves and require significant user adaptation, ultimately reducing the usability in real-world contexts.

Consequently, there remains a clear absence of smart ring systems that support co-located input-output interaction with high sensing granularity, enabling advanced interaction paradigms. Integrating a multi-point touch display into a smart ring addresses this gap by enabling self-contained direct control with immediate on-ring feedback, thereby closing the interaction loop and supporting versatile interaction modalities such as glanceable display and on-the-go interaction.

We identified three key challenges in designing such a smart ring platform: (i) Existing touch display modules [5, 6] provide touch capabilities but are fundamentally incompatible with the geometric constraints of compact wearable form factors such as smart rings. (ii) While prior work on smartphones and smartwatches informs touch and glanceable interaction, smart ring interaction remains underexplored, with limited frameworks, models, and design principles for input interpretation and design. (iii) As evidenced by commercial smart rings [40], assembly is challenging due to tight tolerances, limited space, and curved form factors. Ensuring reproducibility in an open-source platform requires lowering fabrication and modification barriers while maintaining reliability and consistency.

To address this gap, we introduce iRing, an open-source, lightweight, and low-cost smart ring that integrates capacitive multi-touch with a high-resolution pixel display, enabling co-located input and output for one-handed or multi-finger, in-motion interaction. Beyond the hardware integration, we operationalize *nano interactions*: a compact vocabulary of touch primitives adapted for direct, co-located input and visual output on an ultra-small, curved ring display. *Nano interaction* is designed to support compact interaction within a single glance–touch cycle, with input and feedback occurring on the same finger-worn surface. In this paper, we investigate the following research questions:

RQ1: Why is co-located interaction on smart rings needed?

RQ2: What interaction capabilities does co-located interaction on smart rings enable?

RQ3: In which scenarios is co-located interaction on smart rings most effective?

Extensive user study involving 12 users is conducted to evaluate iRing quantitatively and qualitatively. The users evaluate the comfort, likability, and acceptance of iRing. The users also rate the intuitiveness, learning cost, and cognitive pressure of using the **nano interaction**, under different scenarios. Quantitatively, iRing achieved an overall touch recognition accuracy of 97.7% with stable performance across users. In addition, iRing demonstrated high learning efficiency and low user workload, confirmed through standardized and subjective evaluations, and NASA-TLX workload test scoring 2.97 out of 7.0. The detailed evaluation of iRing are presented in Section 5.

In summary, our work makes the following contributions:

- We present iRing, a low-cost, open-source smart ring (Figure 1) centered on a custom capacitive multi-touch display for co-located interaction. We design a transparent flexible PCB (FPCB) enabling pseudo-concurrent multi-point sensing integrated with a flexible 160×32 OLED display. Built with accessible manufacturing methods, including easily producible PCBs and 3D-printed housing, iRing supports reproducibility and broader exploration of on-ring interaction. We further detail the assembly process to facilitate replication, modification, and extension.
- We introduce **nano interactions**, a compact vocabulary that adapts familiar touch primitives—tap, double tap, long press, slide, pinch, and spread—for direct, co-located input and visual output on the curved, ultra-small surface of a smart ring. We further design an on-device, event-driven processing pipeline for iRing that transforms raw capacitive signals into spatial coordinates and temporally compact gesture events, supporting reliable interaction despite the limited sensing area and ambiguity across adjacent electrodes.
- We evaluated iRing and **nano interaction**, quantitatively and qualitatively, to assess system accuracy, user acceptance, and perceived intuitiveness. To further illustrate the interaction space, we developed six representative application scenarios that leverage the interaction techniques and sensing modalities integrated in iRing.

2 Related Work

This section covers two related areas: visual output and interaction techniques in smart rings. We first review prior work on ring-based

Table 1: Summary of related work with output modalities and comparison to iRing.

System	Year	Interaction				Input Sensors	Visual Output	WLS	OS
		Richness	Co-located	On-the-go	Glanceable				
Ketabdar et al. [29]	2012	Low	✓	✓	✗	IMU/Proximity	LED	✗	✗
Bardot et al. [9]	2021	Low	✗	✓	✗	Touchpad	LED	✗	✗
Pradana et al. [41]	2014	Low	✓	✗	✗	Button	RGB LED	✗	✗
Ringly [44]	2015	Low	✓	✓	✗	N/A	RGB LED	✓	✗
Wave for Work [27]	2018	Low	✓	✓	✗	Buttons	Dot-matrix LED	✓	✗
Darbar et al. [15]	2019	Medium	✗	✓	✗	IMU/Touchpad	Rigid Pixel OLED	✓	✗
Zhang et al. [61]	2019	Low	✗	✓	✗	Fluid Sensor	LED	✗	✗
OmniRing [63]	2023	Medium	✗	✓	✗	IMU	N/A	✓	✓
Rogbid SR08 [45]	2024	Medium	✓	✓	✓	Capacitive Button	7-Segment LED	✓	✗
Casio Ring [11]	2024	Low	✓	✓	✓	N/A	7-Segment LCD	✓	✗
ssLOTR [62]	2024	Medium	✗	✓	✗	IMU	N/A	✓	✓
WritingRing [26]	2024	Medium	✗	✓	✗	IMU	N/A	✓	✗
MouseRing [47]	2024	Medium	✗	✓	✗	IMU	N/A	✓	✗
VisRing [33]	2025	Medium	✓	✓	✓	IMU	Flexible Pixel OLED	✓	✓
iRing (ours)	**2026**	High	✓	✓	✓	Multi-point Touchscreen/IMU	Flexible Pixel OLED	✓	✓

visual feedback, including LEDs, segment displays, and pixel-based displays. We then examine ring-based interaction techniques, spanning touch and motion, and discuss their limitations in supporting co-located interaction. Table 1 provides an overview of academic and commercial smart rings most relevant to iRing.

2.1 Visual Output in Smart Ring

Despite prior work on smart rings has investigated various feedback modalities, such as thermal, frictional, and vibrotactile cues, visual feedback [24, 29, 42, 64] has been largely restricted to simple LED indicators, segment LCD displays and, in a few cases, non-flexible displays [13, 15, 41]. Early developments in ring-based visual feedback primarily relied on single or multiple light-emitting diodes (LEDs), as discussed in the works of [13, 29, 41]. Ketabdar et al. [29] introduced a ring equipped with vibrator and a RGB LED to provide combined vibrotactile and visual feedback during gesture interactions with nearby devices. After LEDs, subsequent designs incorporated segment-digit display to present digits and fixed icons. For instance, commercial models such as the Casio CRW-001-1ER [11] and the Rogbid SR08 Ultra [45] employ small 7-segment displays to show numerical information such as time. To enable richer visualizations, later research moved beyond LEDs and segment displays toward pixel-based displays [15, 33, 52]. NailDisplay [52] mounts a miniature screen on the fingernail to provide always-available, fingertip-bound feedback, enabling see-through touch for precise small-target selection, as well as support for imaginary interfaces and mid-air finger interactions. Building on the idea of finger-worn displays, RingIoT [15] integrated infrared (IR) transmitter, IMU sensor, one capacitive sensor, and a rigid OLED display into a finger-worn device for internet-of-things (IoT) interaction, but its bulk and lack of flexibility restricts natural hand and arm movements. More recently, VisRing [33] advanced this direction by introducing a smart ring with a flexible OLED display for miniaturized on-ring visualizations ("nano visualizations"). However, its on-device interaction remained limited, as it does not incorporate a touchscreen for direct, co-located touch interaction.

2.2 Ring-based Interaction

Smart rings have emerged as a promising wearable form factor, offering subtle, always-available interaction on the finger [4, 13, 26, 33, 38]. Early designs primarily support basic input events like buttons, taps, and slide gestures [16, 23, 60]. As [56] reviews more complex modalities, such as IMUs, electromagnetic field sensing, acoustic sensing, cameras, etc., enabling richer gesture recognition, trajectory tracking, text/symbol entry, and direct manipulation. [32, 36, 47, 55]. Prior prototypes rely on external devices such as phones, tablets, or watch screens to deliver visual output cannot provide co-located, immediate, on-ring feedback [30, 38, 43]. However, reliance of external visual feedback requires users to divert their gaze and attention, thereby increasing cognitive effort, as evidenced by the split-attention effect [1]. Recently, several prototypes have integrated LEDs or OLEDs to provide immediate visual feedback [13, 29, 41]. However, they typically convey only simple cues such as colors and strobes, leading to low information bandwidth, limited interactivity, and reduced immersion. To address these research gaps, iRing transforms the ring into a co-located input–output device by integrating a flexible touchscreen with visualization, enables direct access and manipulation on the finger, providing immediate, co-located feedback without diverting attention to external screens, and supports diverse finger-based on-screen interactions such as tap, pinch, and spread.

3 iRing Design

iRing is a touch-display co-located smart ring system that allows users to interact directly with its curved display through touch gestures, enabling real-time, on-device control and glanceable interaction. This section outlines the key challenges in the creation of such a compact wearable device and describes our hardware design process, detailing how we integrated a multi touch sensing subsystem into the ring to enable spontaneous and intuitive interaction.

3.1 Challenges

Designing a compact smart ring with a touch-enabled display for rich interaction presents unique challenges, making it significantly more complex than developing a sensing-only or visualize-only smart ring:

3.1.1 *Designing a Multi-point Touch Screen for Smart Ring.*

Designing a touch screen for iRing poses several unique challenges beyond those encountered in conventional wearable devices. (i) stringent geometric and mechanical constraints imposed by the

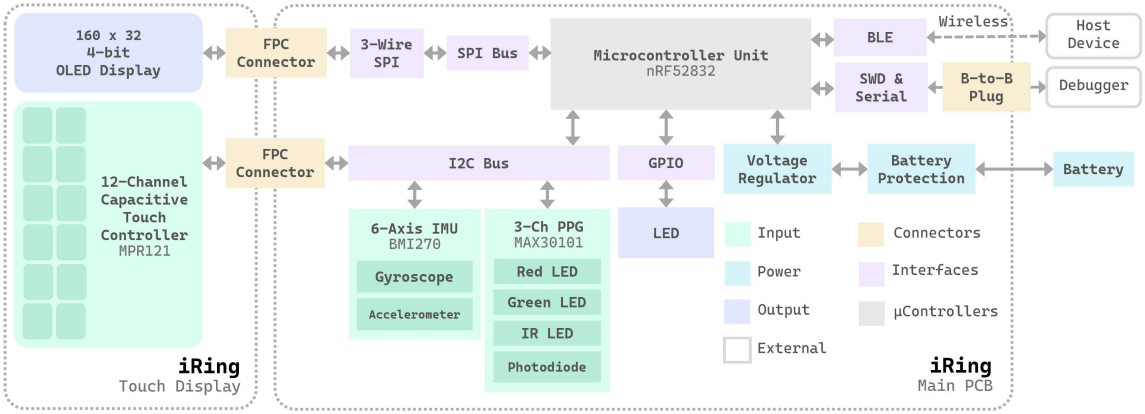


Figure 2: iRing System Block Diagram.

curved, miniaturized form factor; (ii) the need for precise co-design and spatial co-registration between the touch sensor and display to enable accurate co-located interaction; (iii) degraded capacitive sensing performance on flexible, curved substrates, which complicates reliable touch localization and multi-point detection; (iv) fabrication challenges in realizing transparent, flexible electrodes and robust interconnects without compromising durability or sensing fidelity. Hence, designing a touch screen greatly increase the complexity of designing and realizing the iRing platform.

3.1.2 ***Form Factor and Mechanical Design.*** Designing a lightweight, compact touch display smart ring is challenging. (i) extreme planar constraints requiring high-density integration of sensing, power, and wireless modules onto a single flexible PCB; (ii) stringent vertical stack limits (2.4–3.2 mm) that must accommodate multilayer flexible curved display, touch electrodes, and interconnects without compromising wearability; (iii) tight geometric tolerances imposed by the ring's curvature, which are critical to preserving display clarity and touch responsiveness; (iv) the need for a transparent, mechanically robust enclosure that maintains optical transmissivity, sensing performance, and long-term wearing comfort."

3.1.3 ***Manufacturing Feasibility and Cost-effectiveness.*** Selecting materials that balance durability with cost-effictivness under certain structural complexity is also critical to ensure that iRing remains accessible and affordable to the research community for future discoveries. The following challenges should be considered: (i) the need for precise assembly of heterogeneous components—including sensing modules, multilayer flexible PCBs, and a touch-enabled curved display—within a tightly constrained form factor; (ii) integration of the display beneath a transparent, protective enclosure that must simultaneously preserve structural integrity, optical transmittance, and capacitive sensing performance; (iii) fabrication challenges in achieving a clean, durable outer casing without degrading visual clarity or interaction quality; (iv) the requirement to maintain low-cost, scalable processes despite reliance on specialized materials and fabrication techniques."

3.2 Flexible Touch Screen Design

In this section, we present the design of the touch-enabled display for iRing, detailing its sensing modality, underlying interaction principles, and implementation approach.

3.2.1 ***Design Considerations.*** When designing a flexible touchscreen for a smart ring, we consider form factor, resolution, touch sensitivity, power consumption, reliability, and the computational capability of the paired micro-controller to optimize both sensing performance and interaction quality. (i) The form factor must fit the ring's limited space while supporting sufficiently large and dense electrodes for reliable input, and precisely align with the display's active area to enable co-located input–output and direct manipulation without visual occlusion. (ii) The system must provide low-latency response to maintain a tight input–feedback loop, which is critical for preserving the sense of immediacy and control in on-finger interaction. (iii) High temporal resolution is required to support concurrent or pseudo-concurrent sensing, enabling expressive multi-touch gestures and nano interactions beyond discrete tapping. (iv) The touchscreen must be mechanically flexible to conform to the ring geometry without performance degradation, while remaining compatible with integrated components such as circuitry and display. Overall, the design balances physical constraints with interaction demands, enabling responsive, co-located, and expressive touch interaction within the smart ring form factor.

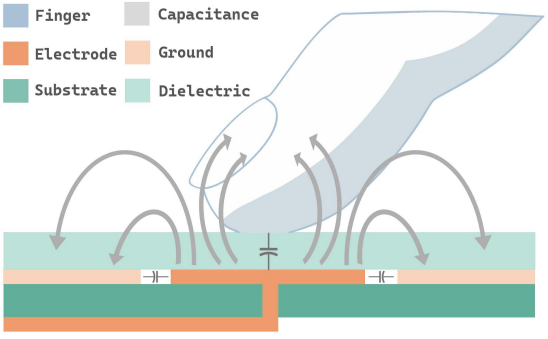


Figure 3: Illustration of Self-Capacitive Sensing.

3.2.2 ***Sensing Technique.*** To achieve the design goals in Section 3.2.1, we adopt multi-touch self-capacitive sensing as the primary input modality for its high sensitivity, fast response, and low-power, cost-effective operation. As shown in Figure 3, each electrode forms a capacitor to ground, and a nearby finger perturbs the electric field, changing its capacitance. By continuously measuring these changes across electrodes, the system estimates touch presence and position. Extending to densely patterned electrodes on

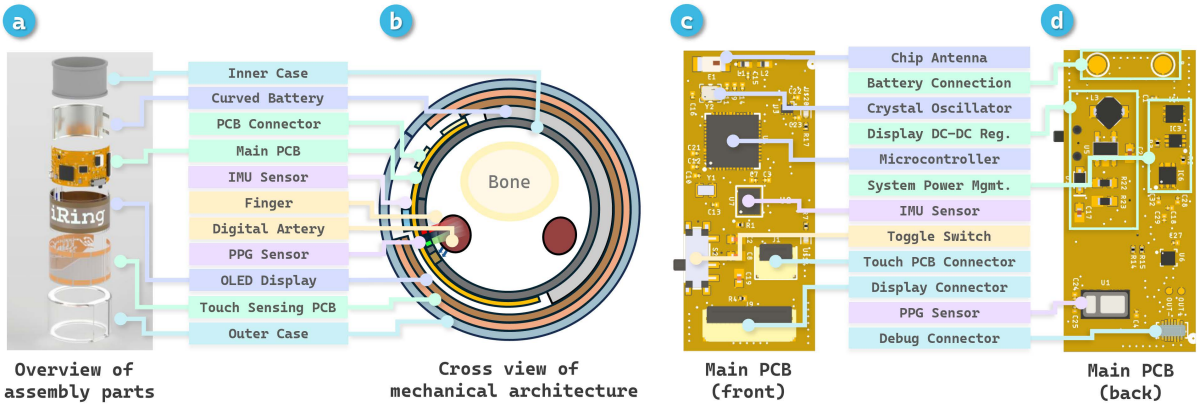


Figure 4: (a) Exploded view of iRing depicting an overview of the components. (b) Cross-section view of iRing (c) The top view of the main PCB board (d) The bottom view of the main PCB board

flexible substrates enables multi-touch sensing and richer gesture dynamics with millimeter-level resolution. The approach supports near real-time operation on resource-constrained micro-controllers with minimal computational overhead and low perceived latency.

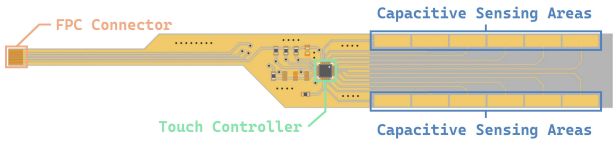


Figure 5: Layout Diagram of the Touch Sensing PCB of iRing.

3.2.3 ***Flexible Touchscreen Circuit Design.*** We present a flexible multi-touch screen that can atop an off-the-shelf flexible OLED display, enabling touch-based interaction on smart rings. We employ NXP's MPR121 [2], a low-latency, low-power self-capacitive controller supporting 12 channels with sampling intervals down to 1 ms and $\sim$ 29 μA active current. As shown in Figure 5, the touch screen FPCB has 12 independent electrodes connected to each channels of the MPR121 controller. We initially explored a traditional opaque FPCB with a display-area cutout; however, it was highly susceptible to unintended activation due to interference from overlapping traces and electrodes. We therefore implement the touchscreen on a two-layer, polyimide-based transparent flexible PCB, enabling trace routing within the display area with minimal visual occlusion. The non-sensing regions are selectively cleared of copper to align with the display's active area, preserving only thin (5mil) traces to ensure unobstructed visualization. This configuration ensures electrical performance under curvature, maintains display transmittance, and simplifies assembly while supporting the ring's structural integrity. The controller interfaces with the MCU via I^2C (up to 400 kHz), enabling sampling intervals of $\sim$ 2 ms per channel. As illustrated in Figures 5, 4c, and 2, the touch PCB is modularized and connects to the main board via an FPC connector, providing a compact, robust, and reliable interface.

3.3 Platform Design

In this section, we present the overall platform design of iRing, covering the touch-enabled display, mechanical design, selected sensing modalities, form factor and circuit configuration, programming interface, and assembly process. We also provide a comprehensive analysis of the manufacturing costs associated with the system.

3.3.1 ***Iteration on Overall Circuit Design.*** In the initial design, we integrated the touch subsystem into the main PCB, resulting in a total length of 134 mm. Given the typical smart ring radius (7–10 mm) [14], this would require wrapping the PCB around the ring two to three times, depending on fit. Our experiments showed that multiple wrappings increase the risk of damaging flexible components, including the display and PCB, and degrade capacitive sensing performance. Degradation also arises from shared ground and power planes between the sensing and main circuitry, introducing ripple and transient noise into the sensing lines [3]. To address this, we separate the system into two PCBs: a 2-layer transparent FPCB for input acquisition (Section 3.2.3) and a 4-layer main PCB for processing and rendering. Details of the main PCB are provided in Section 3.3.3. This separation reduces mechanical stress, improves sensing reliability, and simplifies assembly.

3.3.2 ***Form Factor and Mechanical Design.*** Our goal is to design a lightweight, comfortable, touch-enabled smart ring that supports spontaneous on-screen interaction in daily life. Figures 4a and 4b illustrate the form factor of iRing. To address the dual challenges of integrating multiple ICs within a constrained PCB area and strict mechanical limits (Section 3.1), we combine circuit optimization, structural design, and 3D printing to minimize size while preserving comfort and usability. **Circuit Optimization.** As described in Section 3.3.1, we separate the main PCB and touch-sensing PCB to reduce mechanical fragility and sensing noise, improving space efficiency and enabling a compact form factor. **Structural Design.** As shown in Figure 4b, we organize components—including the touch PCB, main PCB, OLED display, curved battery, and inner/outer casings—into a layered "sandwich" architecture. This arrangement enables compact integration while maintaining mechanical stability and alignment between input and display. **3D Printing.** The device is enclosed in a skin-friendly 3D-printed housing (Figure 6h). A transparent SLA-printed resin outer shell preserves display visibility, while a TPU-printed inner ring provides compliant, size-adaptive fit for different users. iRing weighs 6.2–6.4 g depending on size, with a thickness of 4.05 mm and a width of 18 mm.

3.3.3 ***Main PCB Circuit Design.*** As shown in Figure 4c and 4d, we design a four-layer main PCB that integrates key components, including the PPG sensor, IMU, micro-controller, power management, and display drivers. Assembly details are provided in Section 3.3.4,

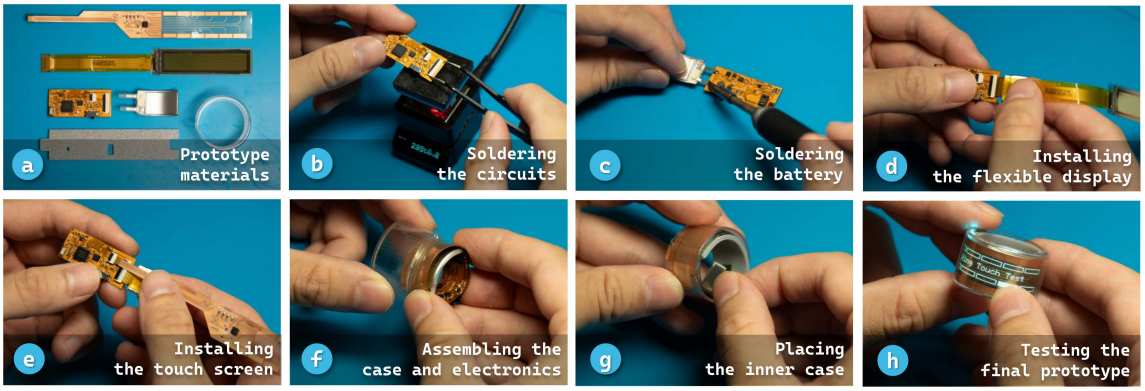


Figure 6: The fabrication and assembly process of the iRing prototype.

and the overall hardware architecture is illustrated in Figure 2. The main PCB is built around an nRF52832 MCU [39] (Nordic Semiconductor), which supports Bluetooth Low Energy communication over 2.4 GHz RF. The MCU communicates with the touch controller, IMU, and PPG sensor via I^2C, and with the OLED display via a 3-wire SPI interface. To accommodate heterogeneous voltage requirements, we design dedicated power regulation circuits. A Texas Instruments TPS71818-33 [54] dual-output regulator supplies 3.3 V to the MCU, IMU, and OLED logic, and 1.8 V to the touch controller and PPG sensor. A DC–DC boost converter (Diodes Inc. AP3012 [17]) generates a 12 V rail to drive the OLED anode. We further incorporate a battery protection system of BQ29700DSER [53] and DMN3032LFDB dual N-MOSFET [18] to prevent overcharge and over-drain. iRing is powered by a curved 60 mAh lithium battery. During active use with sensing, display, and touch interaction, the system draws approximately 9 mA at 3.7 V, enabling up to 6 hours of continuous active operation. Battery life under intermittent everyday use depends on display duty cycle and interaction frequency and is not characterized in the current study.

3.3.4 ***Assembling.*** iRing consists of two FPCBs, a flexible OLED display, an inner case, an outer case, and a curved battery. As described in Sections 3.3.2 and 3.2.3, the touch sensor is implemented on a separate PCB to avoid multiple wrappings, simplifying assembly and preserving capacitive sensing performance. Figure 6 illustrates the step-by-step assembly process. It begins with preparing all components, including the two flexible circuits, curved battery, and OLED display (Figure 6a). Electronic components are first soldered onto the main PCB (Figure 6b–c), followed by connecting the touch-sensing PCB via an FPC connector (Figure 6c). The flexible OLED display is then wrapped around the circuit (Figure 6d–e), after which the assembly and battery are inserted into the outer case (Figure 6f). The inner case is aligned using hand tools (Figure 6g), and the final prototype is tested to verify functionality (Figure 6h).

3.3.5 ***Price Breakdown.*** Table 2 summarizes the retail and wholesale costs of key hardware components and PCB fabrication. Retail prices are based on small batches (5 units), while wholesale prices reflect bulk production (500 units). Firmware and software are open-source, and the 3D-printed enclosure is produced in-house at no additional cost. iRing consists of two PCBs: a four-layer main PCB and a two-layer touch-sensing PCB. The main PCB costs approximately $19.48 (retail) and $2.34 (wholesale), while the touch PCB costs $21.16 (retail) and $3.1(wholesale). Including all components, the total cost per unit is about $120.03 at retail and $46.43 at wholesale, making it more cost-effective than commercial devices such as the Oura Ring [4].

Table 2: Manufacturing Cost Breakdown.

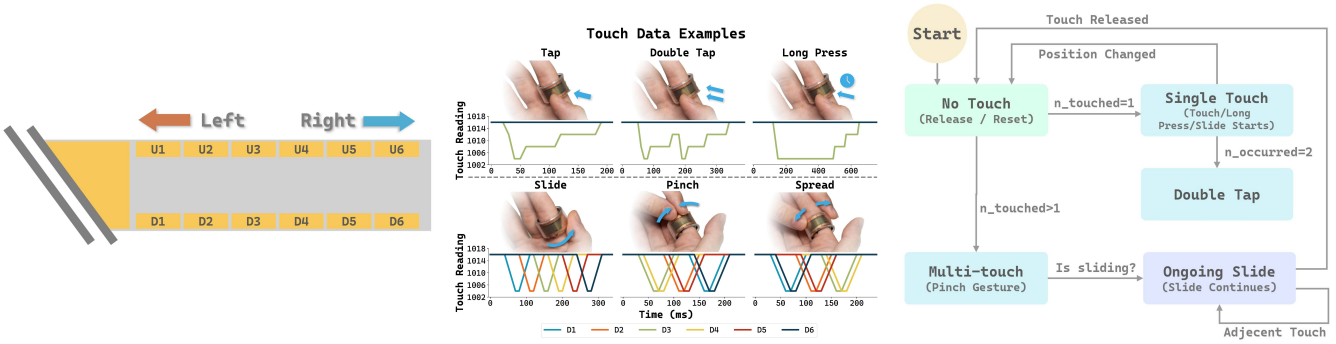
Electronic Component	Unit Price (Retail) [U$]	Unit Price (Wholesale) [U$]
MCU (nRF52832)	5.03	2.39
PPG Sensor (MAX30101)	11.21	6.43
IMU Sensor (BMI270)	3.74	2.57
CAP Sensor (MPR121)	0.58	0.52
OLED Display	37.52	16.94
Chip Antenna	0.59	0.23
Voltage Regulator ICs	1.94	0.94
Battery Protection ICs	0.32	0.18
Others (RCL)	10.45	5.28
Case (Filament)	0.01	0.01
Arc Battery	8.00	7.50
FPCB Manufacturing	40.64	5.44
Total	120.03	≈ 46.43

4 Nano Interaction

In this section, we *nano-interaction*. We articulate the design rationale behind each gesture, grounded in the constraints and affordances of ultra-small, on-body interfaces, and illustrate representative raw touch patterns alongside their mapped interactions. We further detail the underlying algorithms that robustly transform noisy, fine-grained capacitive signals into precise, low-latency touch events in iRing, enabling reliable interaction under the geometric and sensing constraints of the ring-scale surface.

4.1 Input Gesture Types and Sensor Reading

Research prototypes exploring multi-finger gesture control have appeared since the 1980s and 1990s [10, 31, 58], and early commercial devices such as FingerWorks touchpads [20] demonstrated the feasibility of gesture-based input in practice. The subsequent availability of capacitive multi-touch displays on mobile devices in the late 2000s [28] brought these concepts into everyday use. Building on this progression, the **nano interaction** engine adapts six familiar touch primitives—tap, double tap, long press, slide, pinch, and spread—to the curved, ultra-small interaction surface of a smart ring, where limited sensing area, adjacent-electrode activation, touch occlusion, and constrained hand postures shape how



(a) The 2×6 electrode layout (b) Representative electrode responses for the six gestures (c) State machine converts detection into gesture events

Figure 7: Touch sensing and gesture processing in iRing

these primitives can be performed and interpreted. As detailed in Section 3.2, the touch surface was partitioned into 12 regions arranged as a 2×6 grid. For better understanding, we index the upper row as U1-U6 and the lower row as D1-D6 (left-to-right; Figure 7(a)). The MCU polls the touch controller at 50 Hz to retrive all 12 channels of touch data. Figure 7(b) presents the illutration of the six input gestures. The decrease of the reading means a touch has occurred with causing capacitance variation. Simple gestures are defined by their timing: a 'Tap' is short, a 'Double Tap' is two quick taps, and a 'Long Press' is a single, long touch. More complex gestures activate multiple electrodes: a 'Slide' is detected when adjacent touches occur in sequence, while 'Pinch' and 'Spread' are detected when a pair of points are touched at the same time and changes in sequence. To furhter clearify, Figure 7(b) presents the sensor response waveforms when the above gestures are performed over the sensing zone D1 to D6 as illustrated in Figure 7(a).

4.2 Touch Algorithm Design

The sensing pipeline converts per-channel capacitance measurements into gesture events in four stages. The MCU polls all 12 MPR121 [2] channels at 50 Hz and obtains each electrode's filtered reading relative to its tracked baseline. The baseline-relative changes are compared with the configured touch and release thresholds to form a 12-bit activation mask, while their magnitudes are retained to resolve ambiguous adjacent-channel responses. After temporal de-bouncing suppresses brief or spurious transitions, the state machine (Figure 7(c)) combines the number, positions, sequence, and duration of active contacts to generate tap, double-tap, long-press, slide, pinch, and spread events. The algorithm is implemented using a state machine that manages transitions between interaction states, while temporal de-bouncing improves robustness and reduces false positives.

4.3 Touch Gesture State Machine

As shown in Figure 7(c), the algorithm is implemented as a finite state machine with four states: no-touch, single-touch, multi-touch, and ongoing slide. The no-touch state serves as the default, monitoring releases and validating tap events using duration thresholds, ensuring contacts exceed a minimum duration while remaining below the long-press threshold. It also resets internal states for subsequent inputs. In the single-touch state, the algorithm tracks initial contact by initiating timers and recording positions. It detects taps, long presses, and slides using temporal and spatial cues: a long press is recognized when contact persists beyond a timeout without significant movement, while a slide is triggered when positional changes exceed a threshold after de-bouncing. The multi-touch state handles concurrent contacts, focusing on pinch and spread gestures. It computes the distance between the outermost touch points and compares it with the previous frame; a decrease indicates pinch, and an increase indicates spread. Slide detection is suppressed in this state to avoid ambiguity. The ongoing slide state maintains continuous tracking of directional movement, updating slide events based on positional changes relative to previous states. De-bouncing ensures stability and resilience to transient or extraneous touches during motion. State transitions are primarily driven by the number of active contacts, with additional conditions based on timing, positional variation, and internal flags. This design enables fluid adaptation to dynamic interaction while maintaining coherent gesture interpretation. The resulting gesture events and coordinates are passed to visualization functions within the iRing firmware.

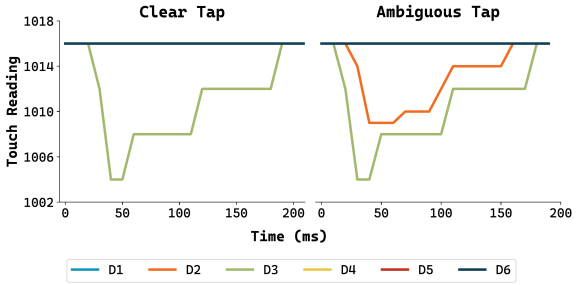


Figure 8: Representative raw electrode responses for clear and ambiguous taps.

4.4 Adjacent-channel Disambiguation

A single finger may overlap two neighboring electrodes, causing both channels to respond and potentially producing a false multi-touch detection. Figure 8 illustrates this behavior. For a clear tap centered on D3, D3 exhibits the dominant decrease from its baseline while the neighboring channels remain stable. When the tap occurs near the boundary between D2 and D3, both electrodes respond, but D3 retains the larger baseline-relative variation. The algorithm therefore groups simultaneously activated adjacent electrodes and

selects the channel with the largest baseline-relative change, $\Delta_i = b_i - r_i$, as the primary touch location, where b_i and r_i denote the baseline and current readings of electrode i, respectively. Spatially separated activation groups are retained as independent contacts, preserving multi-touch input while preventing a broad single-finger contact from being interpreted as a pinch or spread gesture.

5 Evaluation

Our evaluation examines the technical and interaction feasibility of direct touch on a co-located ring display. Beyond recognition accuracy, we assess interaction completion time, learning, perceived workload, usability across activity contexts, and participants' perceptions relative to existing wearable devices. The study provides foundational evidence for the exploration of co-located on-ring interaction.

5.1 Participants

The user study protocol was approved by a university-affiliated Institutional Review Board (IRB), and informed consent was obtained from all participants. We recruited 12 participants (6 males and 6 females, aged 20 to 44, mean = 23.54) from a university campus. All participants had normal or corrected-to-normal vision. Seven had prior experience with smartwatches or smart bands; three had used smart rings without on-device visualization, and one had used a smart ring with on-device visualization. Ring users were not necessarily a subset of smartwatch/band users.

5.2 Procedure and Apparatus

Participants provided informed consent and completed a brief demographics/background questionnaire. They then read an overview describing the gesture set, device/visualization conditions, and the trial flow. Before data collection, each participant completed a 3-minute familiarization with iRing; on-screen prompts and brief coaching ensured they understood the trial procedure. All participants were invited to complete four complementary studies: (1) gesture-class recognition; (2) quantitative evaluation of end-to-end interaction response latency and learning efficiency; (3) baseline comparison against commercial devices; and (4) qualitative assessment of subjective workload and usability under different scenarios. Quantitative sessions were conducted in a uniformly lit, bright, high-ceiling room to minimize illumination variation. Qualitative sessions took place across diverse contexts, including pedestrian settings, cafes, and office spaces. We also recorded observational field notes and participants' spontaneous comments throughout the study.

Study I: Multi-touch Gesture Recognition Performance. This study evaluates iRing's recognition of six gesture classes—tap, double tap, long press, slide, pinch, and spread—and analyzes fine-grained attributes for tap and slide. As shown in Figure 7(a), the touch surface is divided into 12 regions (2×6 grid, U1–U6, D1–D6). For gesture-class recognition, participants completed two sessions of 50 randomized trials (≈10 min each). Each trial displayed a target class (e.g., "Long press"), and accuracy was computed against the prompted class; detected results were shown on iRing for logging. For fine-grained evaluation, each trial specified the full interaction: taps used a single target cell (e.g., "U2"), evenly covering all regions; slides used a same-row start→end pair (e.g., "D5→D2"), encoding start, end, and direction. Participants executed a single continuous slide along the specified row. Ground truth included target location(s) and direction, and predictions were compared accordingly to compute accuracy.

Study II: Quantitative Learning Assessment. To complement the system-level evaluations, we conducted a quantitative user study assessing iRing from an end-user perspective. We examined two dimensions: (i) learning efficiency, i.e., how quickly participants reached stable performance across sessions; and (ii) end-to-end interaction completion time, defined as the total time from on-screen cue to task completion, including action execution, system recognition, and completion feedback. The study covered six gesture-based techniques (Section 4): tap to magnify corresponding numbers (Figure 9a), double tap to toggle between play and stop icons (Figure 9b), long press to complete annular progress ring (Figure 9c), slide to move a slider block from start to destination (Figure 9d), pinch to zoom an alphabet item (Figure 9e), and spread to zoom out an alphabet item (Figure 9f).

Study III: User Qualitative Study. In this study, participants used three applications covering all touch primitives of iRing across different scenarios: media control while running, message scrolling while conversing, and text zooming while working. They also performed the same tasks at rest under full attention. They completed the NASA TLX questionnaire [25] on a laptop to assess subjective workload across six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration. Ratings were collected on a 7-point Likert scale, where higher scores indicate lower workload or higher usability.

Study IV: Baseline Comparison.

To contextualize the perceived usability and interaction niche of iRing, we conduct a qualitative comparison with three representative wearable devices: Fitbit Charge 5 [22], Apple Watch Ultra [7], and the Rogbid SR08 smart ring [45]. Participants performed comparable 5-minute tasks on each device in a fixed order, using only the supported gestures (Section 4). Afterward, they provided subjective ratings on ergonomics, weight, and social acceptability. Participants also used three demo applications spanning one- and two-handed, on-the-go, and glanceable scenarios, including media control while running, message scrolling while conversing, and text zooming while working. These tasks represent one- and two-handed, on-the-go, and glanceable use cases. Participants also performed equivalent tasks on an Apple Watch for comparison. Results were collected as subjective ratings across applications and devices.

5.3 Results

Result I: Multi-touch Gesture Recognition Performance.

Gesture Recognition Accuracy. Figure 10(a) presents the confusion matrices for the six gestures. In the cross-user setting, iRing achieves 97.7% accuracy, indicating robust recognition. Performance is slightly lower for two-finger gestures (pinch, spread) than single-finger gestures (tap, double tap, long press), likely due to higher coordination demands and resolving multiple contacts on a compact surface. Minimal off-diagonal values indicate low confusion and consistent discrimination.

Fine-grained Accuracy of Tap Interaction. Figure 10(b) shows the confusion matrix for taps across 12 regions. Under cross-user

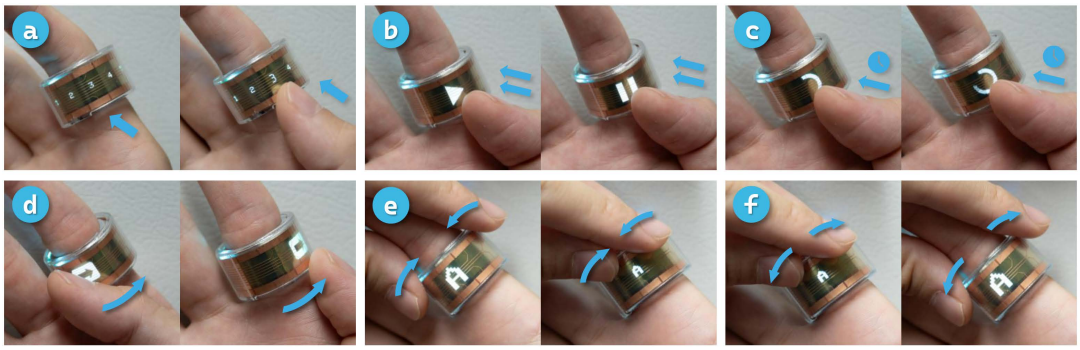
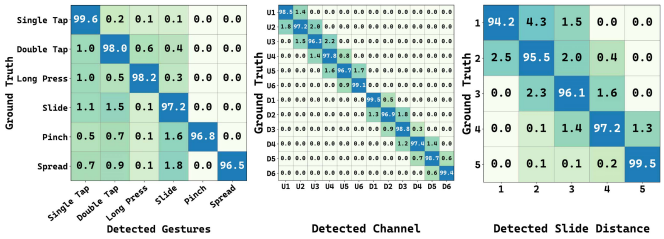


Figure 9: Visual examples of the six interaction primitives performed by participants during the study.



(a) Gesture Accuracy (b) Channel Accuracy (c) Slide Dist. Accuracy

Figure 10: Gesture Recognition Performance.

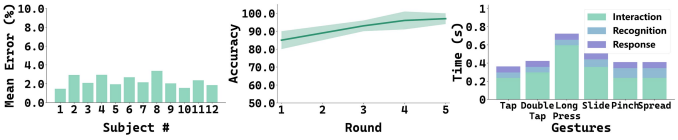
evaluation, iRing achieves 98.0% accuracy, consistent with gesture recognition. Minor errors occur mainly between adjacent regions, likely due to contact overlap in the compact sensing area.

Fine-grained Accuracy of Slide Interaction. Figure 10(c) shows confusion matrices for slides by segment count. The row-normalized matrices exhibit strong diagonal dominance, with errors mainly between adjacent steps. Under cross-user evaluation, iRing achieves 96.5% accuracy, indicating reliable recovery of direction and segment count. Compared to taps, slides are time-extended and more sensitive to speed variation and boundary drift, with occasional interruptions from de-bouncing. Despite this, performance remains stable across directions and segment counts.

Accuracy Across Users. We evaluate cross-user performance. As shown in Figure 11(a), iRing achieves error rates of 1.36%–1.47% across 12 participants, indicating stable generalization, consistent with weak dependence on individual physiology and similar interaction patterns. Lower performance for Subjects 2 and 8 is attributed to smaller finger sizes that limited secure fit, indicating sensitivity to fit quality (e.g., ~8.25 mm radius, size-6 ring [50]). Fabricating reliable prototypes at smaller sizes remains challenging due to display and form-factor constraints, where tighter curvature increases the risk of damaging the flexible OLED. While custom OLEDs could address this, they incur high cost and fabrication complexity. Providing multiple ring sizes is a more practical solution to improve fit, sensing reliability, and interaction accuracy.

Result II: Quantitative Learning Assessment.

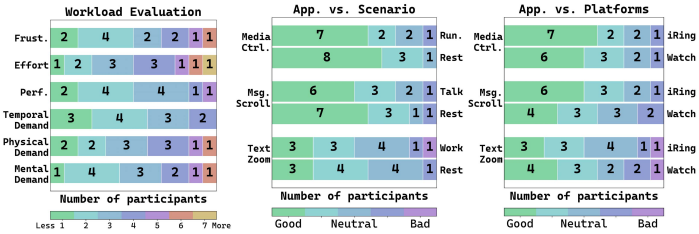
Learning Efficiency. To evaluate learning efficiency, we analyzed task completion time across repeated rounds. As shown in Figure 11(b), accuracy remain stable even from first round, indicating users perform gestures effectively with minimal practice and little improvement over repetition. This suggests that iRing is quickly learnable and supports consistent interaction efficiency.



(a) Accuracy vs. Subject (b) Accuracy vs. Round (c) Latency vs. Gestures

Figure 11: Quantitative Learning Assessment.

Interaction Response Latency. We evaluate interaction latency to assess how quickly iRing recognizes gestures and provides feedback. As shown in Figure 11(c), latency remains low across all gestures, indicating real-time performance. Differences are minimal: tap and slide respond near-instantly, while pinch and spread introduce only slight additional delay. Overall, iRing supports responsive, on-ring feedback essential for fluid interaction.



(a) NASA-TLX Result (b) Experience vs. Scenario (c) Experience vs. Platform

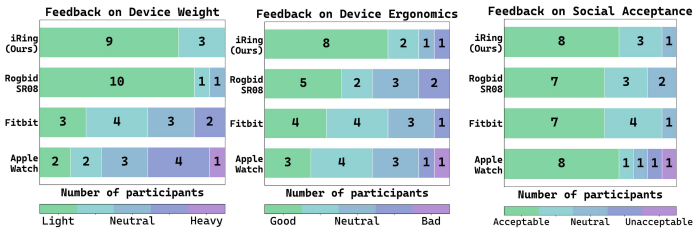
Figure 12: Result of Qualitative Assessment.

Result III: User Qualitative Validation.

NASA-TLX Rating. Figure 12(a) summarizes NASA-TLX results, indicating overall low workload. Frustration averages ~2.9 and physical demand ~2.7, reflecting minimal strain during interaction. Mental demand (~3.1) and temporal demand (~3.0) remain low, suggesting the system is easy to understand and not time-critical. Effort is slightly higher (~3.3), while performance is rated favorably at ~3.5. Overall, these results indicate that iRing supports low cognitive and physical load, suitable for glanceable, on-the-go interaction.

Applications Under Different Scenarios. Figure 12(b) shows user ratings across three applications on iRing under different usage contexts. Ratings in designated scenarios are comparable to those in fully rested, focused conditions, indicating robustness to contextual variations and supporting effective on-the-go interaction and glanceable use in real-world settings.

Result IV: Baseline Comparison. Figure 13 summarizes comparisons between iRing and commercial platforms. As shown in



(a) Feedback of Weight (b) Feedback of Ergonomics (c) Feedback of Acceptance

Figure 13: Baseline Comparison Feedback.

Figure 13(a), iRing achieves competitive weight ratings comparable to the Rogbid SR08. However, unlike iRing, that device lacks a multi-touch, high-resolution display, relying instead on a 7-segment LED and a single capacitive button, limiting interaction expressiveness. iRing also receives high ratings for ergonomics and social acceptance, comparable to existing wearables, indicating positive short-term perceptions of comfort and social acceptability. Shown in Figure 12(c), compare to the Apple Watch, iRing shows higher preference for media control and message scrolling, enabled by one-handed interaction. For the third application, the Apple Watch scores higher, as the digital crown provides more efficient zoom control, although zooming on both devices requires two hands.

Result V: Subjective Feedbacks. At the end of the questionnaire, participants engaged in a discussion to provide feedback on iRing. Several highlighted the novelty of the form factor: *"It feels futuristic to have a touchscreen on a ring—like a mini smartwatch on my finger"* (P11). Others emphasized usability in mobile contexts: *"The one-handed swiping is really convenient; I could see myself using this while walking"* (P1). A majority (11/12) noted the benefit of immediate access: *"I can check information without pulling out my phone or raising my wrist."* Overall, these findings suggest that iRing offers a novel and practical interaction modality through its OLED touchscreen, balancing usability with opportunities for refinement in gesture coordination and power efficiency.

6 Design Implications for Co-located On-ring Interaction

The design and evaluation of iRing reveal that co-located interaction on a smart ring cannot be treated simply as a scaled-down touchscreen. The ring surface is narrow and curved, partially occluded during touch, and constrained by the accessibility of different fingers and hand postures. Its touch sensor and display must also share the same limited surface while maintaining reliable sensing and visible feedback. Drawing from our design process, gesture evaluation, application comparison, and participant feedback, we identify the following considerations for designing co-located input and visual output on ring-scale interfaces.

Balancing Accessibility and Visibility. Co-located input and output compete for the same limited ring surface. Touch regions and feedback should therefore remain accessible and interpretable despite finger occlusion, favoring brief interactions over sustained visual manipulation.

Aligning Input and Feedback. The spatial and temporal structure of visual feedback should correspond directly to detected touch events. In iRing, localized highlighting and brief state changes confirm interaction without requiring to inspect external display.

Designing within Sensing Constraints. Ring-scale gestures should reflect the resolution and ambiguity of the sensing surface. The 2×6 electrode layout therefore favors compact, distinguishable primitives based on contact duration, sequential movement, and relative touch-point separation rather than unrestricted touchscreen input.

Selecting Appropriate Usage Contexts. Co-located ring interaction is most suitable for short tasks requiring immediate, compact feedback. Participants favored iRing for media control and message scrolling, but preferred the smartwatch for zooming, suggesting that ring interaction should complement larger devices according to task duration, visual complexity, and precision requirements.

7 Application Scenarios

We implemented six application scenarios to illustrate the interaction space enabled by co-located touch input and visual output on a smart ring. Three scenarios—media control while running, message scrolling while conversing, and text zooming while working—were included in Studies III and IV, where they were examined across usage contexts and compared with equivalent interactions on an Apple Watch. The remaining scenarios serve as system demonstrations that explore additional uses of the platform. Together, these examples highlight contexts that may benefit from immediate, finger-proximal feedback, while the comparative results also show that larger devices remain preferable for interactions requiring greater visual space or precise control.

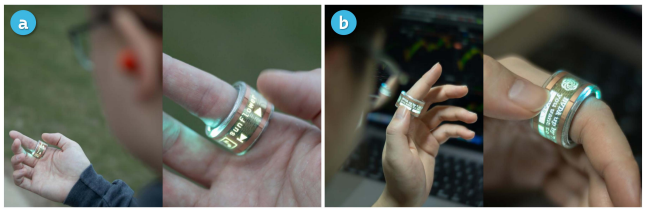


Figure 14: Use Case Examples of iRing.

Media Control During Running In mobile IoT scenarios, interaction often depends on external displays (e.g., smartphone apps), which are not always accessible during movement, limiting usability. In contrast, iRing integrates wireless communication with on-ring, co-located interaction, enabling direct control during running. As shown in Figure 1(c) and Figure 14(a), users can adjust media volume in real time with immediate, glanceable feedback on the ring, supporting low-attention, on-the-go interaction. Beyond media control, iRing can interface with other IoT devices (e.g., lights, thermostats), illustrating how glanceable, co-located feedback could support on-the-go device control.

LLM-mediated Interaction. With the growing adoption of AI-driven assistants, iRing can integrate with large language model (LLM) agents, such as OpenClaw [51], to support intelligent assistance, enabling quick actions and seamless conversational continuity. As shown in Figure 14(b), prompts generated by the LLM are presented directly on the ring, allowing users to select responses and issue commands through co-located interaction. This scenario illustrates how an always-available, glanceable ring interface could support brief interactions with AI services.

Messaging User Interface. To demonstrate iRing's capability for interacting with text elements, we implement a messaging application scenario. As illustrated in Figure 1(c), users can pinch or spread to adjust text size, enabling flexible presentation of dense content within limited display space. iRing further supports one-handed interaction as part of its on-finger form factor: users can use the thumb of the wearing hand to perform sliding gestures on the display for content navigation. The co-located, on-hand design of iRing enables low-attention, on-the-go interaction.

Healthcare. Smart rings are widely used for health and fitness tracking, monitoring heart rate, blood oxygen, and sleep [21, 35, 46]. However, most rely on smartphone apps for interaction, requiring attentional shifts and limiting real-time, on-body use. In contrast, iRing integrates a touch-input flexible curved OLED display, enabling co-located interaction and direct manipulation. Users can slide between health metrics, and the values are updated instantly after measurement, as shown in Figure 1(c). By combining sensing and on-body interaction, iRing provides glanceable access to health data and illustrates how brief health-data interactions may be performed without consulting a secondary display.

Menu Navigation. We explored the design of a menu interface on smart rings. Menu interface is essential when a wearable moves beyond a single-purpose sensor to support interactive functionalities, as it provides a clear entry point for interaction and ensures scalability to future applications. As shown in Figure 1(c), the curved display presents a sequence of application icons that users can scroll through with slide gestures, users can launch the corresponding application by tapping on an icon. We believe that iRing has only scratched the surface of on-screen interaction for smart rings, and diverse applications are to be explored and extended by the research community.

Interactive Gaming. Games provide a compelling scenario for exploring continuous interaction under dynamic visual feedback. iRing provides two key benefits: immediate on-ring visualization which allows players to view game states directly; and an organic interaction modality where users can rotate the ring around the finger for game control, offering a more natural and engaging experience. We implemented the representative Brick Breaker game, as shown in Figure 1(c). In Brick Breaker, users can slide or rotate the ring to control the paddle while monitoring the game state directly on the curved display. This scenario illustrates how touch and ring rotation can be combined with on-device visual feedback for compact game interaction.

8 Limitations and Future Work

8.1 Evaluation Scope and Long-term Use

Our study establishes the short-term feasibility of iRing through a 12-participant evaluation with brief familiarization and controlled or semi-contextual tasks. This scope does not characterize sustained comfort, mechanical durability, accidental activation, fit across broader populations, or everyday adoption. Future work will use contextual benchmarking, longitudinal diary studies, and in-the-wild deployments [8, 12, 49] to examine how users adopt and appropriate co-located ring interaction over time and how iRing functions alongside smartphones, smartwatches, and AR glasses.

8.2 Power Efficiency

Unlike smartwatches, smart rings are constrained by limited battery capacity while supporting multiple features [26, 33], making power efficiency critical. In this work, we explore touch-based interaction within this constraint by tuning display refresh rates to balance usability and energy consumption. Future work will investigate e-ink as a low-power alternative to OLED. Due to its bistable nature, e-ink consumes power only during updates and negligible power when static [19]. For example, a flexible e-ink display draws $\sim 0.01\,\mu$A in standby and ~ 8 mA during refresh [57]; with a 2–4 s refresh cycle [34], a 2 s update interval could support ~ 7.5 hours of use ($\sim$ 13,500 frames). However, e-ink's low refresh rate limits real-time feedback and responsiveness, which is critical for interaction. We therefore retain OLED in the current design. Future efforts will focus on improving e-ink refresh performance and incorporating touch-triggered activation to update the display only during interaction, further extending battery life.

9 Conclusion

We introduced iRing, the first smart ring featuring a flexible multi-touch OLED display with integrated capacitive sensing for on-screen interaction. The prototype weighs 6.2 g and costs approximately 46.43$. We also introduced nano interactions as a compact interaction vocabulary for characterizing and supporting direct, co-located touch and visual feedback on a curved, ultra-small ring surface. To realize this concept, we develope an integrated hardware–software system supporting gesture-based interactions (e.g., tap, slide, pinch), achieving a classification accuracy of 97.7%. We then conducted a user study (N=12) to evaluate usability, comfort, and social acceptance, and to examine the effectiveness and applicability of co-located interaction on a smart ring. Our findings suggest that nano interactions are particularly suitable for brief tasks performed alongside another activity, such as media control while walking or running. Participants' preferences further indicate that iRing can complement smartphones and smartwatches in scenarios that benefit from immediate, finger-proximal feedback, while larger displays remain preferable for interactions requiring sustained visual attention or greater precision. Finally, we illustrated the interaction space of iRing through six application scenarios spanning GUI interaction, IoT control, health monitoring, gaming, and utility navigation. In the future, we will explore adaptive refresh rates and partial updates to improve energy efficiency, as well as context-aware interactions that adapt gestures and feedback to user activity (e.g., walking vs. sitting). These scenarios motivate further exploration of co-located ring interaction for IoT, healthcare, gaming, and immersive environments such as VR/AR.

Acknowledgments

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